

Quick Check

Lesson 4: Building Life's Molecules: Structure of Carbohydrates and Proteins

8 multiple-choice questions • choose A, B, C or D

Sub-Strand 1.2: Chemicals of Life | Lesson 4

Ugali is mostly starch, which is a long chain made of many small units joined together. Which small unit is repeated in a starch chain?

A Amino acid

B Maltose

C Glucose

D Glycogen

Two glucose molecules join to form maltose, a disaccharide. What is the name of the bond that joins them?

A Glycosidic bond

B Peptide bond

C Protein bond

D Amino bond

In a model, each bottle top is an amino acid and each straw is a peptide bond. A group joins 3 bottle tops in a single chain to make a tripeptide. How many straws (peptide bonds) do they need?

A 1

B 3

C 2

D 4

Groups built protein models using bottle tops of different colours for different R-groups. What does this model show about why proteins are so varied?

- A** Different amino acids can be joined in many different orders and combinations
- B** Proteins are made of only one type of amino acid repeated many times
- C** Each protein uses a different kind of bond to join its units
- D** Proteins are varied because they contain more glucose than starch does

Glycogen is stored in animal tissue such as the meat in nyama choma. On a chart with the columns Monosaccharide, Disaccharide, Polysaccharide and Protein, where does glycogen belong?

A Monosaccharide

B Polysaccharide

C Protein

D Disaccharide

Glucose contains carbon, hydrogen and oxygen only. Which extra element is found in amino acids but NOT in glucose?

A Oxygen

B Hydrogen

C Nitrogen

D Carbon

A child from Kisumu eats only ugali every day and develops a swollen belly and thin limbs. Which explanation is best?

- A** Ugali contains no glucose, so the child has no energy at all
- B** Githeri contains more starch than ugali, so it prevents kwashiorkor
- C** Ugali is not enough in quantity, and the child would be healthy if given more ugali
- D** Ugali gives glucose for energy but no amino acids, so the body cannot build proteins such as muscle and albumin

Which pair correctly matches a molecule with its main function in the body?

- A** Haemoglobin: energy storage
- B** Haemoglobin: a transport protein that carries oxygen
- C** Glucose: transports oxygen in the blood
- D** Glycogen: body building and repair

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).